

The practice of category theory

In the actual practice of doing category theory here are some things that often come up.

Morphisms

The whole starting point of category theory is to have morphisms between objects, not just objects by themselves. This idea pervades category theory by us always thinking about maps between structures, not just individual structures. So we think about totalities of a particular type of structure, not just one example of the structure. With morphisms to hand, we think about isomorphisms instead of equalities, and more generally the weakest form of sameness for the dimension we’re in.

Reasoning

We use diagrammatic reasoning to make use of geometric intuition as well as logical and algebraic intuition. Proofs are often essentially type-checking. We invoke the “obvious axioms” which might seem obscure but once you understand the principle it’s less obscure: the principle is that if we have two inherently structural ways of mediating between the same things, they should be the same in order to prevent structural incoherence.

Coherence

We often have two approaches to the same structure, one that starts locally and expands to a global view, and one that starts globally and then pins down a local view that suffices. These two approaches are summarized in the following table.

	Approach 1	Approach 2
definition:	small set of conditions	broad structure
theorem:	broad structure follows	small set of conditions suffices

The question of coherence is the question of how the local view and the global view correspond to one another. Structures in category theory usually rely on having both, and understanding the relationship between them.

Universal properties

We try and characterize things by properties rather than intrinsic characteristics, and best of all we characterize by universal property, where possible. This

sometimes involves souping up a category in order to find a way to express something as a simple universal property in a complicated category, and then unraveling it to be a complicated universal property in a more fundamental category.

Structural awareness

We maintain an awareness of our interaction with structure and how we're using it. For example if we are making use of associativity we make sure we're aware of it rather than taking it for granted. This helps for generalizations, especially into higher dimensions where axioms that we used to take for granted may become structural isomorphisms that we now have to keep track of. We also maintain awareness of whether we're invoking the *existence* of some structure, or exhibiting a *specific* piece of structure. For example, we remain aware of whether we're saying things are "isomorphic", or whether we're exhibiting a specific isomorphism between them. Being specific is harder but gives us a more precise understanding.

Higher dimensions for nuance

Morphisms are higher-dimensional than objects and give us more nuance. More generally, every higher dimension we involve gives us even more nuance, along with more complications. But the aim is to learn how to deal with the complications in order to benefit from the added nuance. I think this is important in life as well. Arguments in life have become too black-and-white. I wish we could all become more higher-dimensionally nuanced in our arguments in math and in every part of life.

APPENDICES

